

EKA Portal: System Modernization & Redesign Project Roadmap

This document outlines the step-by-step master plan to modernize the digital infrastructure of the Greek Community of Alexandria (EKA). It combines the technical stability requirements from "PROJECT PROPOSAL: EKA PORTAL PHP CORE MODERNIZATION 2026-06-03.pdf" and the visual/functional enhancements from "The Greek Community of Alexandria (EKA) CMS update Proposal 2024-10-12.pdf" into a single, risk-free execution strategy.

Phase 1: Local Sandbox Environment & Infrastructure

Objective: Build an exact copy of the website on a local laptop to work safely without risking downtime on the live production server.

- **Task 1.1: Local Server Initialization**
 - Set up a localized Nginx web server and database engine (MariaDB/MySQL) on the local workstation.
- **Task 1.2: Database Snapshot Extraction**
 - Export and download the core 400+ MB database ledger from the live server.
- **Task 1.3: Media Storage Routing (The 60GB Footprint Optimization)**
 - Configure a specialized Nginx network proxy hack. This allows the local test site to read the massive 60+ GB media library directly from the live web without copying files locally, saving significant disk space and setup time.
- **Task 1.4: Local Domain Mapping (backstage.ekalexandria.org)**
 - Route the staging domain locally on the workstation and run initial database search-and-replace scripts to ensure all internal links function correctly in isolation.

Phase 2: Technical Debt Audit & Plugin Rationalization

Objective: Identify obsolete code structures and eliminate heavy legacy plugins that degrade performance or cause active 500 server errors.

- **Task 2.1: Legacy Plugin Analysis**
 - Audit all currently active plugins to trace functionality, performance bottlenecks, and structural compatibility with modern PHP standards.
- **Task 2.2: Decommissioning Ledger (The "Throw-Away" List)**
 - Isolate bloated layout builders, legacy sliders, and redundant tracking utilities that can be permanently removed or replaced by native, lightweight block elements.
- **Task 2.3: Core Feature Extraction Mapping**
 - Document the unique, essential behaviors of the plugins marked for removal (e.g., custom data fields or multilingual logic) to prepare for clean reimplementation.

Phase 3: Legacy Layout Extraction & Replication (Pivot)

Objective: As design approvals for modernization have been pending since June 28, we are proceeding with a strict 1:1 visual replication of the existing website's layout. This unblocks the critical technical upgrades while maintaining the familiar aesthetic.

- ☐ **Task 3.1: Style Extraction & Variable Mapping**
 - Extract typography, colors, and layout widths from the live ekalexandria.org site to establish the foundation for theme.json and block settings.
- ☐ **Task 3.2: Legacy Structural Replication**
 - Map existing Homepage, Internal Pages, and News/Category layouts to native block patterns without altering the original visual aesthetic.
- ☐ **Task 3.3: Specialty Page Development**
 - Implement structural layouts for the Tachydrómos Newsletter archive and Board of Directors pages within the legacy visual constraints.
- ☐ **Task 3.4: Third-Party Plugin Mitigation**

- Replace legacy embedded elements (like LayerSlider) with modern, lightweight equivalents (e.g., Carousel Block) that visually match the original design.

Phase 4: Greenfield Theme Development (The Flagship Block Core)

Objective: Build a custom, high-purity, modern WordPress theme from absolute scratch using strict engineering standards.

- ☐ **Task 4.1: Structural Theme Scaffolding**
 - Initialize a completely blank, lightweight, block-native theme architecture (theme.json) completely independent of legacy visual builders.
- ☐ **Task 4.2: SCSS Style Architecture Setup**
 - Establish a clean, compiled style ecosystem (SCSS) to govern typography scales, spacing tokens, and color profiles globally without inline code bloat.
- **Task 4.3: High-Performance Page Templates**
 - Translate approved visual designs into optimized block layouts for the homepage, inner content, and operational views.

Phase 5: Re-Engineering Missing Features & Integrations

Objective: Program clean, native replacements for the essential features of the legacy plugins we threw away.

- **Task 5.1: Tachydromos Archive Engine Customization**
 - Code clean, fast database filters and sorting functions to display the newsletter archive natively.
- **Task 5.2: Board of Directors Engineering**
 - Build custom FSE page templates and queries to dynamically map and display translated board members across three languages natively.
- ☐ **Task 5.3: News Section Optimization**
 - Build optimized query loops to surface high-priority community news dynamically.
- ☐ **Task 5.4: Modern Secure Token Setup (Mailchimp Engine)**
 - Develop the direct OAuth token loop to securely pass communications to external marketing list engines without utilizing vulnerable text passwords.
- ☐ **Task 5.5: Community Advertisement Injection Slots**
 - Program clean layout positions for relevant community advertisements to safely run without tracking scripts slowing down page loading.

Phase 6: Full Data Migration & System Calibration

Objective: Merge the historical database content with the brand-new theme and modern database tables.

- **Task 6.1: Legacy Data Transformation Scripting**
 - Execute targeted script operations to map old post formats into clean, standardized block patterns (e.g., migrating legacy shortcode grids into native Gutenberg blocks and CPTs for Newsletters and Board Members).
- ☐ **Task 6.2: Database Cleanup & Schema Alignment**
 - Purge historical plugin remnants, clear junk metadata, and optimize database indexing across the 400+ MB file ledger to stabilize table performance.
- ☐ **Task 6.3: Multi-Language Validation Drill**
 - Stress-test the language switching integrity for all historic posts across English, Greek, and Arabic configurations.

Phase 7: Production Cutover & Final Launch

Objective: Deploy the modernized platform to the live production server with zero downtime.

- ☐ **Task 7.1: Server Runtime Hardening**
 - Configure the live production stack environment to modern PHP standards.
- ☐ **Task 7.2: Final Content Sync Delta**

- Run a final differential database migration to capture any new articles or announcements published on the live portal during local development.

- ☐ **Task 7.3: Deployment, Routing & Upstream Flushes**

- Push the new theme and cleaned database live, point local Nginx configurations to the local media assets folder, clear system caches, and verify complete global site availability.